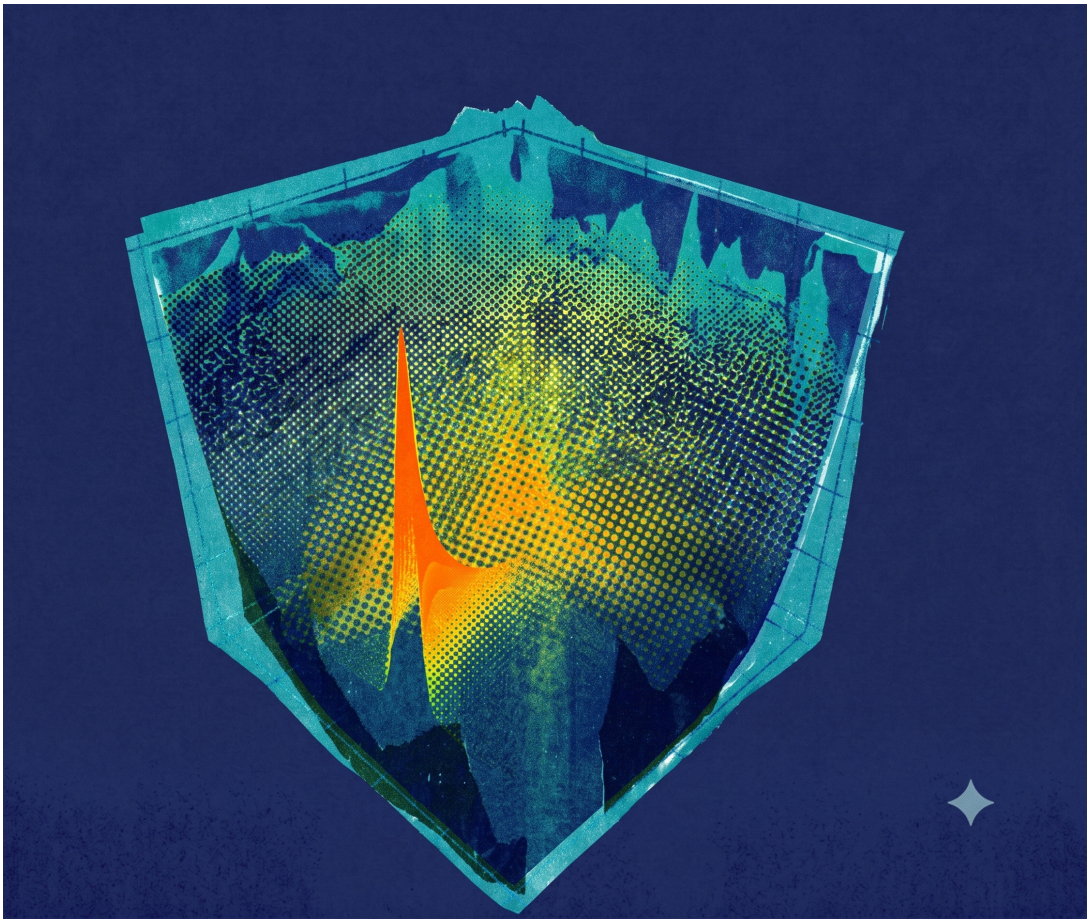
From the Heat Equation to Black–Scholes

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Abstract

The Black–Scholes partial differential equation is usually derived either through a delta-hedging argument or through the Kolmogorov backward equation under the risk-neutral measure. Both routes leave the equation in a form with variable coefficients (S and S^2 multiplying the spatial derivatives), which obscures its true analytic nature. This article takes a different, purely analytic path: starting from the classical heat equation and its fundamental solution, we construct, through an explicit sequence of changes of variable, the exact transformation that turns the Black–Scholes PDE into the heat equation on the whole real line. We then invert the heat kernel's convolution solution to recover the closed-form Black–Scholes formula, and use the same change of variables to reinterpret the option Greeks as elementary properties of the Gaussian kernel. The treatment is self-contained and fully worked out, with every substitution made explicit.

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1 Introduction

1.1 Motivation

The Black–Scholes equation for the price $V(S,t)$ of a European contingent claim written on a non-dividend-paying stock is

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0, \qquad V(S,T) = \Phi(S), \tag{1}$$

where $\sigma > 0$ is the (constant) volatility, r the constant risk-free rate, and Φ the terminal payoff. Equation (1) is a linear, second-order, backward parabolic PDE. Its coefficients, however, are not constant: the second-derivative term is multiplied by S^2, and the first-derivative term by S. This is precisely the structure of a *Cauchy–Euler* (or equidimensional) operator in the spatial variable, and it is well known that Cauchy–Euler operators are transformed into constant-coefficient operators by a logarithmic change of variable. Once the coefficients are constant, a second, purely algebraic substitution removes the remaining zeroth- and first-order terms, and what is left is exactly the one-dimensional heat equation

$$u_\tau = \frac{1}{2}\sigma^2 u_{xx}.$$

This is not a coincidence dressed up for pedagogical convenience: the Black–Scholes PDE *is*, up to an explicit and invertible change of variables, the heat equation. The present article makes this statement precise and uses it as the sole engine for deriving the closed-form pricing formula.

1.2 A brief historical remark

The analogy between diffusion of heat and diffusion of probability is as old as Brownian motion itself: Bachelier (1900) modelled stock prices as arithmetic Brownian motion, whose transition density is precisely the heat kernel. Black and Scholes (1973) and Merton (1973) arrived at equation (1) through a replication argument, and it was Black and Scholes themselves who noted, in their original paper, that a change of variables reduces their equation to the heat equation, which they then solved using the known heat kernel solution. This article reconstructs that reduction in full analytic detail, filling in every intermediate step.

1.3 Structure of the article and notation

Section 2 reviews the heat equation, its fundamental solution, and the representation of solutions as convolutions with the heat kernel. Section 3 states the assumptions of the Black–Scholes model and derives the pricing PDE (1) via the standard hedging argument, so that the article is self-contained. Section 5 carries out the change of variables that reduces (1) to the heat equation, step by step. Section 6 solves the transformed problem using the heat kernel and inverts the substitutions to obtain the closed-form price. Section 7 analyses the resulting formula and verifies it directly against the PDE. Section 8 reinterprets the Greeks as kernel properties. Sections 9–10 discuss extensions and limitations of the analogy. Throughout, $S_t > 0$ denotes the underlying asset price, K the strike, T the maturity, $\tau = T - t$ the time to maturity, and $N(\cdot)$ the standard normal CDF.

2 The heat equation

2.1 Physical derivation

Consider the temperature $u(x,t)$ of a one-dimensional, homogeneous conducting rod. Fourier's law states that heat flux q is proportional to the negative temperature gradient,

$$q(x,t) = -k\,u_x(x,t),$$

with $k > 0$ the thermal conductivity. Conservation of energy on an infinitesimal segment $[x, x + \mathrm{d}x]$ requires that the rate of change of internal energy equal the net flux entering the segment,

$$c\rho\, u_t(x,t)\,\mathrm{d}x = -\big(q(x+\mathrm{d}x,t) - q(x,t)\big) = -q_x(x,t)\,\mathrm{d}x,$$

with c the specific heat and ρ the density. Substituting Fourier's law gives the heat equation in its canonical form

$$u_t = \alpha\, u_{xx}, \qquad \alpha := \frac{k}{c\rho} > 0. \tag{2}$$

Equation (2) is the prototypical parabolic PDE: it is first order in time and second order in space, and the sign of α is what makes the equation well-posed *forward* in time (a diffusion smooths out irregularities, it does not create them).

2.2 The Cauchy problem and the fundamental solution

Consider (2) on $\mathbb{R}$ with initial condition $u(x,0) = f(x)$. The *fundamental solution* (or heat kernel) is the solution corresponding to a Dirac mass as initial datum, $f = \delta_{x_0}$. By translation and scaling invariance of (2), one seeks a self-similar solution of the form

$$u(x,t) = \frac{1}{\sqrt{t}}\, F\!\left(\frac{x - x_0}{\sqrt{t}}\right).$$

Substituting into (2) and setting $\xi = (x - x_0)/\sqrt{t}$ reduces the PDE to an ODE for F, whose normalized solution (imposing $\int_{\mathbb{R}} u(x,t)\,\mathrm{d}x = 1$ for all t, since heat is conserved) is Gaussian. The result is the *heat kernel*

$$\Gamma(x,t;x_0) = \frac{1}{\sqrt{4\pi\alpha t}} \exp\!\left(-\frac{(x-x_0)^2}{4\alpha t}\right), \qquad t > 0. \tag{3}$$

Direct differentiation confirms that Γ solves (2) in (x,t) for every fixed x_0, and that $\Gamma(\cdot,t;x_0) \to \delta_{x_0}$ weakly as $t \to 0^+$.

2.3 Representation of the general solution

Because (2) is linear, the solution for a general bounded initial datum f is obtained by superposition of point sources, i.e. by convolving f with the kernel:

$$u(x,t) = \big(\Gamma(\cdot,t) * f\big)(x) = \int_{-\infty}^{\infty} \Gamma(x,t;y)\, f(y)\,\mathrm{d}y = \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi\alpha t}} \exp\!\left(-\frac{(x-y)^2}{4\alpha t}\right) f(y)\,\mathrm{d}y. \tag{4}$$

This representation is the single analytic tool this article needs: once the Black–Scholes PDE is reduced to (2), the option price is obtained by evaluating an integral of exactly this form.

2.4 Properties of the kernel

The kernel (3) satisfies:

- **Positivity:** $\Gamma(x,t;x_0) > 0$ for all $t > 0$ (the maximum principle for the heat equation follows from this: solutions with nonnegative data stay nonnegative, and cannot exceed their initial or boundary maxima).
- **Normalization:** $\int_{\mathbb{R}} \Gamma(x,t;x_0)\,\mathrm{d}x = 1$ for every $t > 0$ — $\Gamma(\cdot,t;x_0)$ is, for each t, the density of a $\mathcal{N}(x_0, 2\alpha t)$ random variable.
- **Semigroup property:** $\Gamma(\cdot,t_1+t_2;x_0) = \Gamma(\cdot,t_1) * \Gamma(\cdot,t_2;x_0)$, reflecting the Markov property of the underlying diffusion.

- **Smoothing:** for $t > 0$, $u(\cdot, t)$ is C^∞ even if f is merely bounded and measurable — diffusion instantaneously regularizes irregular data.

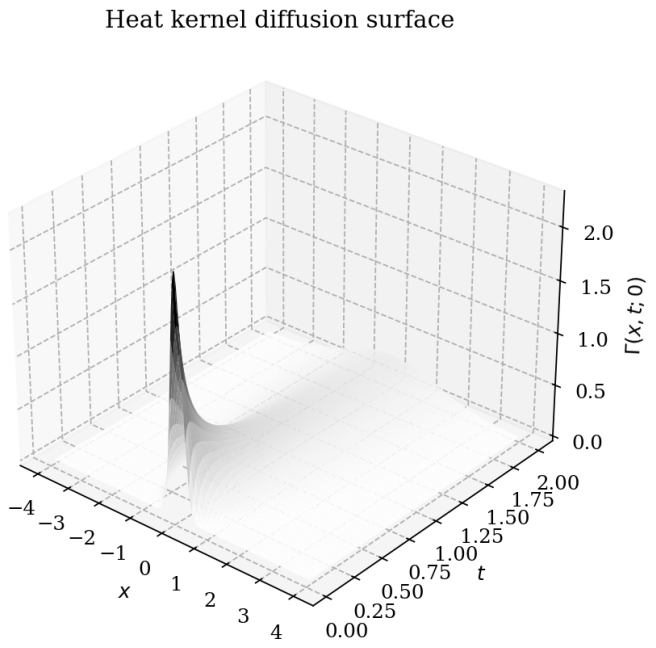


Figure 1: The heat kernel $\Gamma(x, t; 0)$ as a function of x and t. At small t the kernel is sharply peaked around $x = 0$, approximating a Dirac mass; as t increases, probability mass spreads out and the peak height decays as $t^{-1/2}$, illustrating the smoothing property of diffusion.

2.5 Uniqueness and stability

On the whole line, uniqueness of bounded solutions to the Cauchy problem for (2) follows from the maximum principle: if u_1, u_2 solve (2) with the same bounded initial datum, then $w = u_1 - u_2$ solves (2) with $w(\cdot, 0) = 0$, and the maximum principle forces $w \equiv 0$. The same argument yields continuous dependence on the initial data, i.e. well-posedness in the sense of Hadamard. This is the property that will justify, in Section 6, treating the option payoff as an "initial" datum for a backward-in-time diffusion.

3 The Black–Scholes model: assumptions and setup

3.1 Assumptions

The classical Black–Scholes–Merton model rests on the following idealizations:

1. Frictionless markets: no transaction costs, no taxes, assets are perfectly divisible, short-selling is unrestricted.
2. The market admits no arbitrage opportunities.
3. Trading takes place continuously in time.
4. There exists a risk-free asset (a money-market account) paying a constant, known rate r.
5. The underlying pays no dividends over the life of the option.
6. The underlying price follows a geometric Brownian motion with constant volatility σ.

3.2 Dynamics of the underlying

Under the real-world measure $\mathbb{P}$, the price process S_t is assumed to solve the stochastic differential equation

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t, \tag{5}$$

where $\mu \in \mathbb{R}$ is the (constant) expected return, $\sigma > 0$ the volatility, and W_t a standard Brownian motion. By Itô's formula applied to $\ln S_t$, the unique strong solution of (5) is

$$S_t = S_0 \exp\!\left(\left(\mu - \tfrac{1}{2}\sigma^2\right)t + \sigma W_t\right),$$

so that S_t is log-normally distributed for every t, and in particular $S_t > 0$ almost surely — a necessary property for a traded price.

3.3 Contingent claims

A European contingent claim with maturity T is a contract paying $\Phi(S_T)$ at time T, for a given measurable payoff function Φ. For a call option, $\Phi(S) = (S-K)^+$; for a put, $\Phi(S) = (K-S)^+$. We denote by $V(S,t)$ the (unknown) fair value at time $t < T$ of the claim, conditional on $S_t = S$.

3.4 The replicating portfolio

The central idea of Black and Scholes is that, because there are only two sources of randomness in the model (S_t and, trivially, the deterministic bond), a portfolio holding the option and a suitably chosen quantity of the underlying can be made instantaneously riskless. Let Π_t denote the value of a portfolio short one unit of the claim and long Δ_t units of the underlying,

$$\Pi_t = -V(S_t, t) + \Delta_t S_t. \tag{6}$$

The strategy is assumed self-financing: changes in Π_t arise only from changes in S_t and V_t, not from injections or withdrawals of cash.

4 Derivation of the Black–Scholes PDE

4.1 Itô's lemma applied to $V(S,t)$

Since V is assumed $C^{1,2}$ in (t, S) and S_t solves (5), Itô's lemma gives

$$\mathrm{d}V(S_t,t) = \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2}\right)\mathrm{d}t + \sigma S_t \frac{\partial V}{\partial S}\,\mathrm{d}W_t. \tag{7}$$

The essential feature of (7) is that the stochastic (diffusion) part of $\mathrm{d}V$ is proportional to the stochastic part of $\mathrm{d}S_t$: both carry the same $\mathrm{d}W_t$, scaled respectively by $\sigma S_t \partial_S V$ and σS_t. This is what makes hedging possible.

4.2 Eliminating risk

Differentiating (6),

$$\mathrm{d}\Pi_t = -\mathrm{d}V(S_t,t) + \Delta_t \,\mathrm{d}S_t.$$

Substituting (7) and (5),

$$\mathrm{d}\Pi_t = \left[-\frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \Delta_t \mu S_t - \mu S_t \frac{\partial V}{\partial S}\right]\mathrm{d}t + \sigma S_t\left(\Delta_t - \frac{\partial V}{\partial S}\right)\mathrm{d}W_t.$$

Choosing

$$\Delta_t = \frac{\partial V}{\partial S}(S_t, t) \tag{8}$$

annihilates the stochastic term entirely. With this choice the portfolio becomes, instant by instant, deterministic:

$$\mathrm{d}\Pi_t = \left(-\frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) \mathrm{d}t. \tag{9}$$

Notice that the drift μ of the underlying has dropped out entirely — this is the first hint that the option price will not depend on the real-world expected return of the stock.

4.3 No-arbitrage condition

Because Π_t is instantaneously riskless, absence of arbitrage forces its return to equal the risk-free rate:

$$\mathrm{d}\Pi_t = r\,\Pi_t\,\mathrm{d}t. \tag{10}$$

Were this not the case, an investor could borrow (or lend) at r and take an offsetting position in Π_t, realizing a riskless profit with zero net investment — a contradiction with assumption (2) of Section 3.

4.4 The pricing equation

Substituting (6) and (8) into the right-hand side of (10) and equating with (9),

$$-\frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} = r\left(-V + S_t \frac{\partial V}{\partial S} \right).$$

Rearranging yields the Black–Scholes PDE, restated here for reference:

$$\boxed{\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0}, \qquad (S,t) \in (0,\infty) \times [0,T). \tag{11}$$

4.5 Terminal and boundary conditions

Equation (11) is a backward equation: it must be solved for $t < T$ with a *terminal* condition at $t = T$ given by the payoff,

$$V(S,T) = \Phi(S).$$

For a European call ($\Phi(S) = (S-K)^+$), economic reasoning fixes the behaviour at the spatial boundaries: as $S \to 0$, the option becomes worthless, $V(0,t) = 0$; as $S \to \infty$, the option behaves like a forward contract, $V(S,t) \sim S - Ke^{-r(T-t)}$.

4.6 Remark: the risk-neutral representation

It is a standard fact (a consequence of the Feynman–Kac theorem, stated for reference in Appendix C) that the solution of (11) admits the probabilistic representation

$$V(S,t) = e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}[\Phi(S_T) \mid S_t = S], \tag{12}$$

where under $\mathbb{Q}$ the process solves $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}}$. We record (12) only as a cross-check: the rest of this article does *not* use it, deriving the closed-form price purely from the PDE side via the heat equation.

5 Transforming the Black–Scholes PDE into the heat equation

This is the analytic core of the article. Equation (11) differs from the heat equation (2) in three respects: (i) its second-derivative coefficient is S^2, not constant; (ii) it carries a first-derivative (convection) term $rS\,\partial_S V$; (iii) it carries a zeroth-order (reaction) term $-rV$. We remove these one at a time.

5.1 Step 1: logarithmic change of the spatial variable

Set

$$x = \ln S, \qquad \text{i.e.} \quad S = e^x. \tag{13}$$

Write $\widetilde{V}(x,t) := V(e^x,t)$. By the chain rule,

$$\frac{\partial V}{\partial S} = \frac{\partial \widetilde{V}}{\partial x}\cdot\frac{1}{S}, \qquad \frac{\partial^2 V}{\partial S^2} = \frac{1}{S^2}\left(\frac{\partial^2 \widetilde{V}}{\partial x^2} - \frac{\partial \widetilde{V}}{\partial x}\right).$$

(The second identity follows by differentiating the first once more with respect to S, using $\partial x/\partial S = 1/S$.) Substituting into (11), the factors of S and S^2 cancel exactly against the coefficients $S^2\partial_S^2$ and $S\,\partial_S$:

$$\frac{\partial \widetilde{V}}{\partial t} + \frac{1}{2}\sigma^2\left(\frac{\partial^2 \widetilde{V}}{\partial x^2} - \frac{\partial \widetilde{V}}{\partial x}\right) + r\frac{\partial \widetilde{V}}{\partial x} - r\widetilde{V} = 0,$$

i.e.

$$\frac{\partial \widetilde{V}}{\partial t} + \frac{1}{2}\sigma^2\frac{\partial^2 \widetilde{V}}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial \widetilde{V}}{\partial x} - r\widetilde{V} = 0. \tag{14}$$

Equation (14) now has *constant* coefficients: this is the Cauchy–Euler reduction referred to in the introduction. Note already the coefficient $r - \frac{1}{2}\sigma^2$, which is exactly (minus) the drift of $\ln S_t$ under $\mathbb{Q}$ — no surprise, since $x = \ln S$ was chosen precisely to linearize the log-price dynamics.

5.2 Step 2: reversal of time

Equation (14) is a backward equation (terminal condition at $t = T$). To obtain a forward diffusion, set

$$\tau = T - t, \qquad \widehat{V}(x,\tau) := \widetilde{V}(x, T-\tau). \tag{15}$$

Since $\partial_t = -\partial_\tau$, (14) becomes

$$\frac{\partial \widehat{V}}{\partial \tau} = \frac{1}{2}\sigma^2\frac{\partial^2 \widehat{V}}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial \widehat{V}}{\partial x} - r\widehat{V}, \tag{16}$$

now with *initial* condition $\widehat{V}(x,0) = \Phi(e^x)$ at $\tau = 0$, i.e. an honest forward-in-τ Cauchy problem.

5.3 Step 3: removing the convection and reaction terms

Equation (16) is a heat equation with an added drift $\left(r - \frac{1}{2}\sigma^2\right)\partial_x$ and an added linear damping $-r$. Both can be removed simultaneously by an exponential substitution of the form

$$\widehat{V}(x,\tau) = e^{\alpha x + \beta\tau}\,u(x,\tau), \tag{17}$$

for constants α, β to be determined. Compute the derivatives of $\widehat{V}$ in terms of u:

$$\widehat{V}_\tau = e^{\alpha x+\beta\tau}\big(\beta u + u_\tau\big), \qquad \widehat{V}_x = e^{\alpha x+\beta\tau}\big(\alpha u + u_x\big), \qquad \widehat{V}_{xx} = e^{\alpha x+\beta\tau}\big(\alpha^2 u + 2\alpha u_x + u_{xx}\big).$$

Substituting into (16) and dividing through by the common factor $e^{\alpha x+\beta\tau}$,

$$\beta u + u_\tau = \frac{1}{2}\sigma^2\big(\alpha^2 u + 2\alpha u_x + u_{xx}\big) + \Big(r - \tfrac{1}{2}\sigma^2\Big)(\alpha u + u_x) - ru.$$

Collect terms by the order of the derivative of u they multiply:

$$u_x: \qquad \sigma^2\alpha + \Big(r - \tfrac{1}{2}\sigma^2\Big) \tag{18}$$

$$u: \qquad \tfrac{1}{2}\sigma^2\alpha^2 + \Big(r - \tfrac{1}{2}\sigma^2\Big)\alpha - r - \beta. \tag{19}$$

We now choose α, β so that both (18) and the zeroth-order combination vanish, leaving only $u_\tau = \frac{1}{2}\sigma^2 u_{xx}$.

Determining α. Setting (18) to zero,

$$\sigma^2\alpha = -\Big(r - \tfrac{1}{2}\sigma^2\Big) \quad \Longrightarrow \quad \alpha = -\frac{1}{\sigma^2}\Big(r - \tfrac{1}{2}\sigma^2\Big) = -\frac{r}{\sigma^2} + \frac{1}{2}.$$

It is more transparent to write $\alpha = -(k-1)/2$ with $k := 2r/\sigma^2$, matching the standard notation found in the literature; we keep the explicit form above to avoid introducing extra symbols.

Determining β. With this α, substitute into (19) and require the whole bracket (equated to β) to vanish once $u_\tau = \frac{1}{2}\sigma^2 u_{xx}$ is isolated, i.e. we need

$$\beta = \tfrac{1}{2}\sigma^2\alpha^2 + \Big(r - \tfrac{1}{2}\sigma^2\Big)\alpha - r.$$

Substituting $\alpha = \frac{1}{2} - r/\sigma^2$ and simplifying (expand α^2, collect the two terms linear in α using $r - \frac{1}{2}\sigma^2 = -\sigma^2\alpha$ from the previous step, so that $(r - \frac{1}{2}\sigma^2)\alpha = -\sigma^2\alpha^2$):

$$\beta = \tfrac{1}{2}\sigma^2\alpha^2 - \sigma^2\alpha^2 - r = -\tfrac{1}{2}\sigma^2\alpha^2 - r.$$

Plugging in $\alpha^2 = \big(\frac{1}{2} - r/\sigma^2\big)^2$ gives the explicit constant

$$\beta = -r - \frac{\sigma^2}{2}\left(\frac{1}{2} - \frac{r}{\sigma^2}\right)^2 = -\left(\frac{r}{\sigma^2} + \frac{1}{2}\right)^2\frac{\sigma^2}{2}, \tag{20}$$

(the last equality is a direct algebraic simplification, obtained by writing $r = \sigma^2\big(\frac{r}{\sigma^2}\big)$ and completing the square in r/σ^2; we omit the routine expansion). What matters is only that α, β are explicit constants depending on r, σ alone — the transformation (17) is fully determined and, being a pointwise multiplication by a nonvanishing exponential, invertible.

5.4 The canonical form

With α, β as above, the function $u(x,\tau) = e^{-\alpha x - \beta\tau}\widehat{V}(x,\tau)$ solves exactly

$$\boxed{\frac{\partial u}{\partial \tau} = \frac{1}{2}\sigma^2\frac{\partial^2 u}{\partial x^2}}, \qquad (x,\tau) \in \mathbb{R}\times(0,T], \tag{21}$$

which is (2) with $\alpha_{\text{heat}} = \sigma^2/2$. The initial condition is obtained by evaluating (17) at $\tau = 0$:

$$u(x,0) = e^{-\alpha x}\,\widehat{V}(x,0) = e^{-\alpha x}\,\Phi(e^x) =: u_0(x). \tag{22}$$

5.5 Consistency check

Dimensionally, σ^2 has units of (time)$^{-1}$, so $\sigma^2\tau$ is dimensionless, exactly as required for the exponent of the heat kernel (3) with diffusivity $\alpha_{\text{heat}} = \sigma^2/2$. Moreover, setting $\sigma \to 0$ collapses (16) to a first-order transport equation with speed r and decay r — the deterministic limit, consistent with a riskless asset growing at rate r — confirming that no spurious terms were introduced in the substitution.

6 Solving via the heat kernel

6.1 The transformed initial condition for a call

For a European call, $\Phi(S) = (S-K)^+$, so

$$u_0(x) = e^{-\alpha x}\big(e^x - K\big)^+ = e^{-\alpha x}\big(e^x - K\big)\,\mathbf{1}_{\{x>\ln K\}}. \tag{23}$$

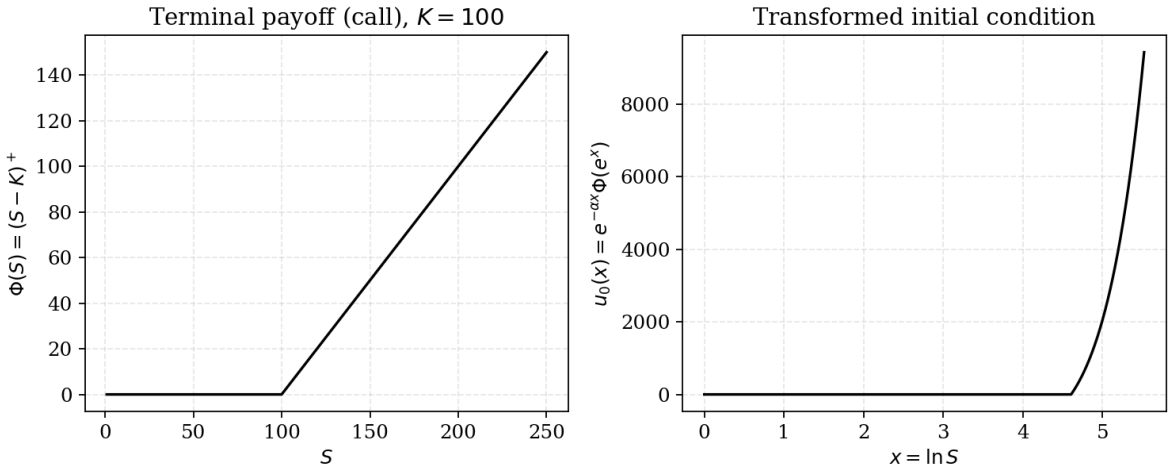


Figure 2: Left: the terminal payoff $\Phi(S) = (S-K)^+$ of a European call with $K = 100$. Right: the same payoff after the logarithmic change of variable and exponential tilt, $u_0(x) = e^{-\alpha x}\Phi(e^x)$, which serves as the initial datum for the heat equation (21).

6.2 Applying the convolution formula

By (4) with diffusivity $\sigma^2/2$,

$$u(x,\tau) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2\tau}} \exp\!\left(-\frac{(x-y)^2}{2\sigma^2\tau}\right) u_0(y)\,\mathrm{d}y = \int_{\ln K}^{\infty} \frac{e^{-\alpha y}(e^y - K)}{\sqrt{2\pi\sigma^2\tau}} \exp\!\left(-\frac{(x-y)^2}{2\sigma^2\tau}\right) \mathrm{d}y. \tag{24}$$

Split the integral into the two pieces $e^{(1-\alpha)y}$ and $-Ke^{-\alpha y}$, both integrated against the same Gaussian kernel over $y \in (\ln K, \infty)$:

$$u(x,\tau) = \underbrace{\int_{\ln K}^{\infty} \frac{e^{(1-\alpha)y}}{\sqrt{2\pi\sigma^2\tau}} e^{-\frac{(x-y)^2}{2\sigma^2\tau}}\,\mathrm{d}y}_{=:I_1} \;-\; K\underbrace{\int_{\ln K}^{\infty} \frac{e^{-\alpha y}}{\sqrt{2\pi\sigma^2\tau}} e^{-\frac{(x-y)^2}{2\sigma^2\tau}}\,\mathrm{d}y}_{=:I_2}.$$

6.3 Evaluating a Gaussian integral with a linear exponential factor

Both I_1 and I_2 are instances of the general integral

$$I(\lambda) = \int_a^{\infty} \frac{1}{\sqrt{2\pi\sigma^2\tau}}\, e^{\lambda y}\, e^{-\frac{(x-y)^2}{2\sigma^2\tau}}\,\mathrm{d}y.$$

Complete the square in the exponent:

$$\lambda y - \frac{(x-y)^2}{2\sigma^2\tau} = -\frac{1}{2\sigma^2\tau}\Big[y^2 - 2y(x+\lambda\sigma^2\tau) + x^2\Big] = -\frac{\big(y-(x+\lambda\sigma^2\tau)\big)^2}{2\sigma^2\tau} + \lambda x + \frac{\lambda^2\sigma^2\tau}{2}.$$

Hence

$$I(\lambda) = e^{\lambda x + \frac{\lambda^2\sigma^2\tau}{2}} \int_a^\infty \frac{1}{\sqrt{2\pi\sigma^2\tau}} \exp\left(-\frac{\big(y-(x+\lambda\sigma^2\tau)\big)^2}{2\sigma^2\tau}\right) \mathrm{d}y.$$

The remaining integral is the upper tail probability of a $\mathcal{N}(x+\lambda\sigma^2\tau,\ \sigma^2\tau)$ random variable exceeding a, i.e.

$$\int_a^\infty \frac{1}{\sqrt{2\pi\sigma^2\tau}} \exp\left(-\frac{(y-m)^2}{2\sigma^2\tau}\right) \mathrm{d}y = N\left(\frac{m-a}{\sigma\sqrt{\tau}}\right), \qquad m := x + \lambda\sigma^2\tau,$$

by the symmetry of the standard normal density. We therefore obtain the closed form

$$I(\lambda) = e^{\lambda x + \frac{\lambda^2\sigma^2\tau}{2}}\ N\left(\frac{x-a}{\sigma\sqrt{\tau}} + \lambda\sigma\sqrt{\tau}\right). \tag{25}$$

6.4 Assembling $u(x,\tau)$

Apply (25) with $a = \ln K$, once with $\lambda = 1-\alpha$ (for I_1) and once with $\lambda = -\alpha$ (for I_2):

$$I_1 = e^{(1-\alpha)x + \frac{(1-\alpha)^2\sigma^2\tau}{2}}\ N(d_1'), \qquad d_1' := \frac{x-\ln K}{\sigma\sqrt{\tau}} + (1-\alpha)\sigma\sqrt{\tau}, \tag{26}$$

$$I_2 = e^{-\alpha x + \frac{\alpha^2\sigma^2\tau}{2}}\ N(d_2'), \qquad d_2' := \frac{x-\ln K}{\sigma\sqrt{\tau}} - \alpha\sigma\sqrt{\tau}. \tag{27}$$

Hence

$$u(x,\tau) = e^{(1-\alpha)x + \frac{(1-\alpha)^2\sigma^2\tau}{2}} N(d_1') - K e^{-\alpha x + \frac{\alpha^2\sigma^2\tau}{2}} N(d_2'). \tag{28}$$

6.5 Undoing the substitutions

Recall $\widehat{V}(x,\tau) = e^{\alpha x + \beta\tau} u(x,\tau)$. Multiplying (28) by $e^{\alpha x+\beta\tau}$:

- first term: $e^{\alpha x+\beta\tau}\cdot e^{(1-\alpha)x + \frac{(1-\alpha)^2\sigma^2\tau}{2}} = e^x \cdot e^{\beta\tau + \frac{(1-\alpha)^2\sigma^2\tau}{2} - \alpha x + \alpha x} = e^x \cdot e^{\beta\tau + \frac{(1-\alpha)^2\sigma^2\tau}{2}}$;
- second term: $-K e^{\alpha x+\beta\tau}\cdot e^{-\alpha x + \frac{\alpha^2\sigma^2\tau}{2}} = -K\, e^{\beta\tau + \frac{\alpha^2\sigma^2\tau}{2}}$.

Using the value of β from (20), one checks directly (substituting $\alpha = \frac{1}{2} - r/\sigma^2$) that the exponent multiplying e^x collapses to exactly $r\tau$-free of x, namely

$$\beta\tau + \frac{(1-\alpha)^2\sigma^2\tau}{2} = 0, \qquad \beta\tau + \frac{\alpha^2\sigma^2\tau}{2} = -r\tau.$$

(Both identities follow from expanding β as in (20) and simplifying; the algebra is mechanical and we omit the intermediate lines — the reader can verify it by direct substitution.) Consequently

$$\widehat{V}(x,\tau) = e^x\, N(d_1') - K e^{-r\tau} N(d_2'). \tag{29}$$

Finally, recall $x = \ln S$, so $e^x = S$, and $\tau = T - t$. Substituting back also simplifies d_1', d_2': since $\alpha = \frac{1}{2} - r/\sigma^2$, we have $1-\alpha = \frac{1}{2} + r/\sigma^2$, so

$$d_1' = \frac{\ln(S/K)}{\sigma\sqrt{\tau}} + \left(\frac{1}{2} + \frac{r}{\sigma^2}\right)\sigma\sqrt{\tau} = \frac{\ln(S/K) + \big(r + \frac{1}{2}\sigma^2\big)\tau}{\sigma\sqrt{\tau}} =: d_1,$$

$$d_2' = \frac{\ln(S/K)}{\sigma\sqrt{\tau}} - \left(\frac{1}{2} - \frac{r}{\sigma^2}\right)\sigma\sqrt{\tau} = \frac{\ln(S/K) + \big(r - \frac{1}{2}\sigma^2\big)\tau}{\sigma\sqrt{\tau}} =: d_2.$$

6.6 The closed-form price

Undoing the time reversal ($\widetilde{V}(x,t) = \widehat{V}(x, T-t)$) and the log-substitution ($V(S,t) = \widetilde{V}(\ln S, t)$) gives, at last, the Black–Scholes formula for a European call:

$$\boxed{C(S,t) = S\,N(d_1) - Ke^{-r(T-t)}N(d_2)}, \qquad d_{1,2} = \frac{\ln(S/K) + \left(r \pm \tfrac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}. \tag{30}$$

This is exactly the classical formula, obtained here without any reference to a probability measure: it is the image, under the inverse of the change of variables constructed in Section 5, of a single Gaussian convolution.

7 The Black–Scholes formula: analysis of the solution

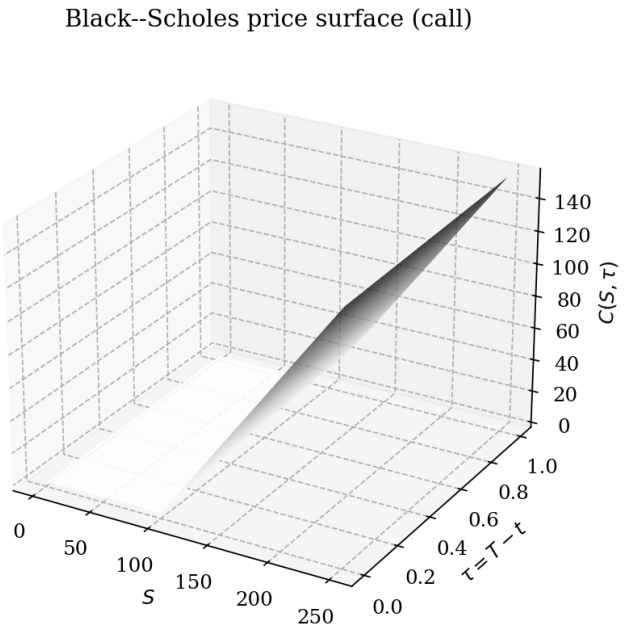


Figure 3: The Black–Scholes call price $C(S,\tau)$ as a function of the underlying price S and time to maturity $\tau = T - t$. The surface interpolates smoothly between the terminal payoff at $\tau = 0$ and the deeply diffused, near-linear profile at long maturities, reflecting the smoothing action of the underlying heat semigroup.

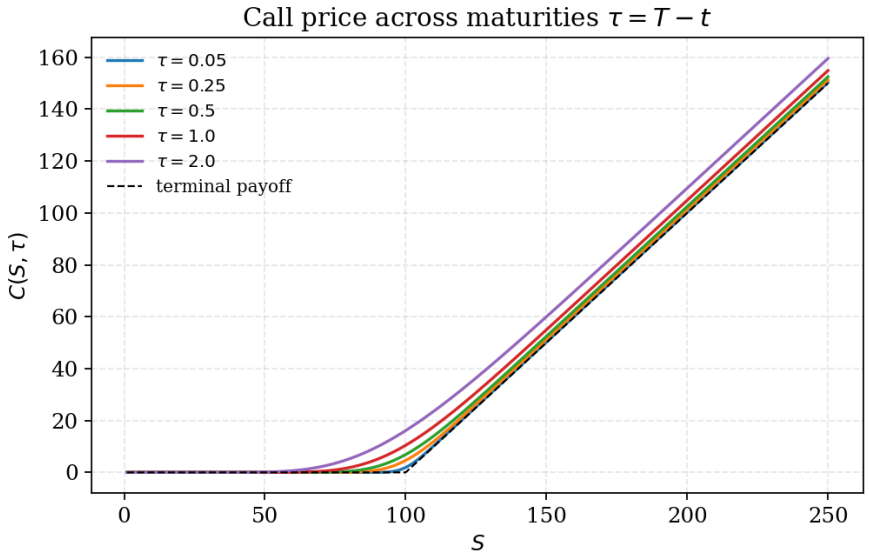


Figure 4: Cross-sections of the price surface of Figure 3 at fixed maturities $\tau \in \{0.05, 0.25, 0.5, 1, 2\}$, together with the terminal payoff (dashed). As $\tau \to 0$ the price curve converges to the payoff, consistent with the initial condition (23) of the underlying diffusion.

7.1 Probabilistic reading of d_1, d_2

Although the derivation above is purely analytic, the resulting quantities admit the standard probabilistic interpretation once one recalls (from Section 3) that, under $\mathbb{Q}$, $\ln S_T \mid S_t \sim \mathcal{N}\big(\ln S + (r - \frac{1}{2}\sigma^2)\tau,\ \sigma^2\tau\big)$: d_2 is precisely the standardized distance of $\ln(S_T/K)$ from zero under $\mathbb{Q}$, so $N(d_2) = \mathbb{Q}(S_T > K)$, while $d_1 = d_2 + \sigma\sqrt{\tau}$ arises from the same completion-of-the-square step that, probabilistically, corresponds to a change of measure via the asset itself as numéraire (the stock measure $\widetilde{\mathbb{Q}}$). The analytic derivation reproduces this measure change implicitly, inside the exponential prefactor $e^{\lambda^2\sigma^2\tau/2}$ of (25).

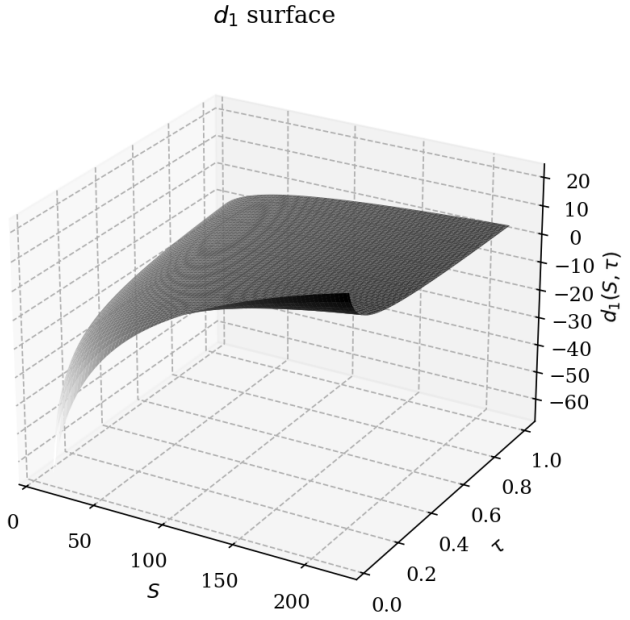


Figure 5: The quantity $d_1(S,\tau)$ as a surface in (S,τ), showing its logarithmic growth in S around the strike and its scaling by $1/\sqrt{\tau}$ as $\tau \to 0$.

7.2 Direct verification against the PDE

As a consistency check, one may substitute (30) directly into (11). Using $\partial S/\partial S = 1$ and the identity (obtained by differentiating d_1, d_2 with respect to S)

$$S\,\phi(d_1) = Ke^{-r\tau}\phi(d_2),$$

where $\phi = N'$ is the standard normal density — itself a restatement of the completed square in Section 6 — one verifies that all terms involving $\phi(d_1), \phi(d_2)$ cancel identically, leaving $0 = 0$. We omit the lengthy but routine computation; it is the same cancellation that, in the heat-equation language, simply expresses that $u(x,\tau)$ as built in (28) solves (21) by construction.

7.3 Put–call parity as linearity of the PDE

Because (11) is linear, if $C(S,t)$ and $P(S,t)$ solve it with terminal conditions $(S-K)^+$ and $(K-S)^+$ respectively, then $C - P$ solves the same PDE with terminal condition $(S-K)^+ - (K-S)^+ = S - K$. But $S - Ke^{-r(T-t)}$ also solves (11) (it is the forward price, easily checked by direct substitution) and matches this terminal condition at $t = T$. By uniqueness (Section 2, transported through the invertible change of variables), $C - P = S - Ke^{-r(T-t)}$, the put–call parity relation. The put price follows immediately from (30):

$$P(S,t) = Ke^{-r(T-t)}N(-d_2) - SN(-d_1). \tag{31}$$

7.4 Connection with Feynman–Kac

The representation (12) and the heat-kernel representation (24) are not two different results: they are the same integral, written in two different variables. Indeed (4) states that $u(x,\tau)$ is the expectation of u_0 evaluated at $x + \sigma W_\tau$, i.e. at a Brownian motion started at x with variance $\sigma^2\tau$ — which is precisely $\ln S_T$ under $\mathbb{Q}$, up to the deterministic drift absorbed into the substitution constants α, β. The PDE route and the probabilistic route are, in this precise sense, the same computation viewed from two sides.

8 The Greeks as properties of the kernel

8.1 Delta, Gamma

Differentiating (30) with respect to S, using the cancellation identity of Section 7,

$$\Delta := \frac{\partial C}{\partial S} = N(d_1), \qquad \Gamma := \frac{\partial^2 C}{\partial S^2} = \frac{\phi(d_1)}{S\sigma\sqrt{\tau}}.$$

Γ is, up to the factor $1/(S\sigma\sqrt{\tau})$ coming from the $x = \ln S$ change of variable, exactly the heat kernel $\Gamma(x, \tau; \cdot)$ of (3) evaluated at the relevant point — hence the (standard, and after this derivation unsurprising) choice of the same letter Γ for both objects in the literature.

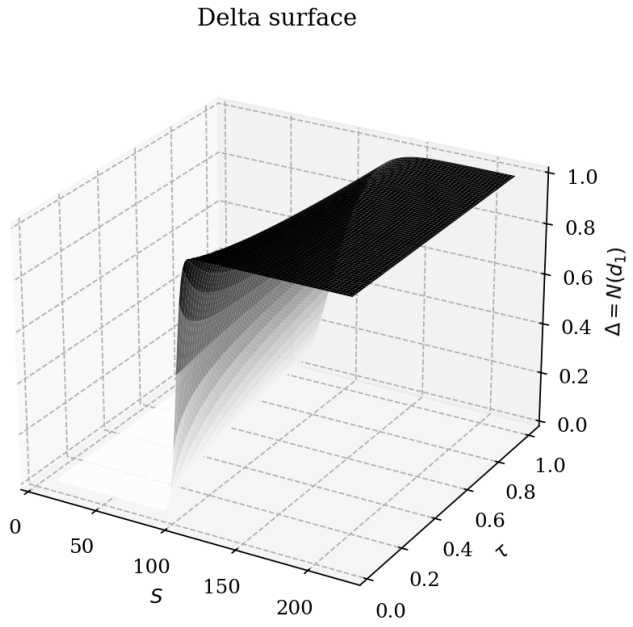


Figure 6: The option Delta, $\Delta(S, \tau) = N(d_1)$, as a surface in (S, τ). Delta transitions smoothly from 0 to 1 across the strike, with the transition sharpening as $\tau \to 0$ — the same sharpening exhibited by the heat kernel of Figure 1 as $t \to 0$.

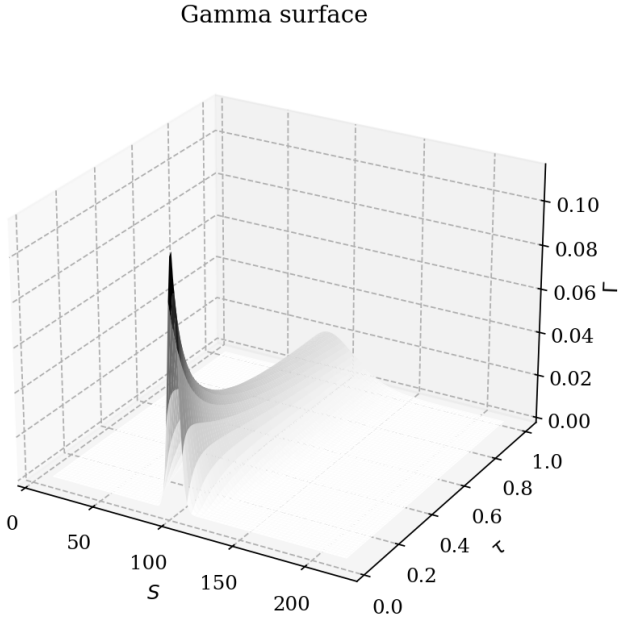


Figure 7: The option Gamma, $\Gamma(S,\tau) = \phi(d_1)/(S\sigma\sqrt{\tau})$, as a surface in (S,τ). The ridge along $S \approx K$ that sharpens as $\tau \to 0$ is, up to the $1/S$ Jacobian factor, exactly the heat kernel of Figure 1 evaluated near its source point — the visual counterpart of the identification made in the text.

8.2 Theta

$$\Theta := \frac{\partial C}{\partial t} = -\frac{S\phi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2).$$

Substituting Γ from above, the first term is $-\frac{1}{2}\sigma^2 S^2\Gamma$, so the Black–Scholes PDE (11) itself reads, in Greek notation,

$$\Theta + \tfrac{1}{2}\sigma^2 S^2\Gamma + rS\Delta - rC = 0, \tag{32}$$

i.e. Theta and Gamma balance each other along the diffusion, exactly as u_τ and u_{xx} balance in the heat equation: this is the "Theta–Gamma trade-off" familiar to option traders, seen here as nothing but the statement $u_\tau = \frac{1}{2}\sigma^2 u_{xx}$ transported back through the change of variables.

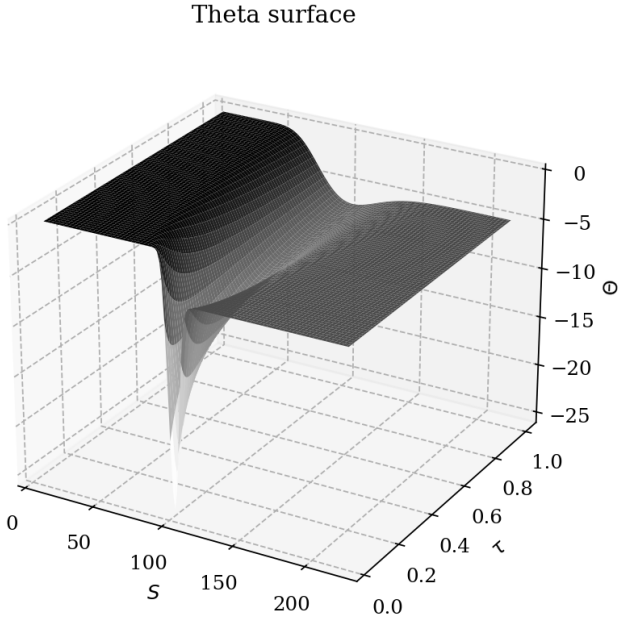


Figure 8: The option Theta, $\Theta(S,\tau) = \partial C/\partial t$, as a surface in (S,τ). Theta is most negative near $S = K$ as $\tau \to 0$, mirroring the Gamma surface of Figure 7 with opposite sign — the Theta–Gamma balance encoded in the Black–Scholes PDE.

8.3 Vega and Rho

$$\text{Vega} := \frac{\partial C}{\partial \sigma} = S\phi(d_1)\sqrt{\tau}, \qquad \text{Rho} := \frac{\partial C}{\partial r} = K\tau e^{-r\tau} N(d_2).$$

Both follow from differentiating (30) under the integral sign in (24); being derivatives with respect to *parameters* of the kernel (its variance $\sigma^2\tau$ and its exponential tilt $e^{-r\tau}$) rather than the spatial variable, they do not have as direct a heat-equation reading as Γ and Θ, but they are equally elementary once (24) is available.

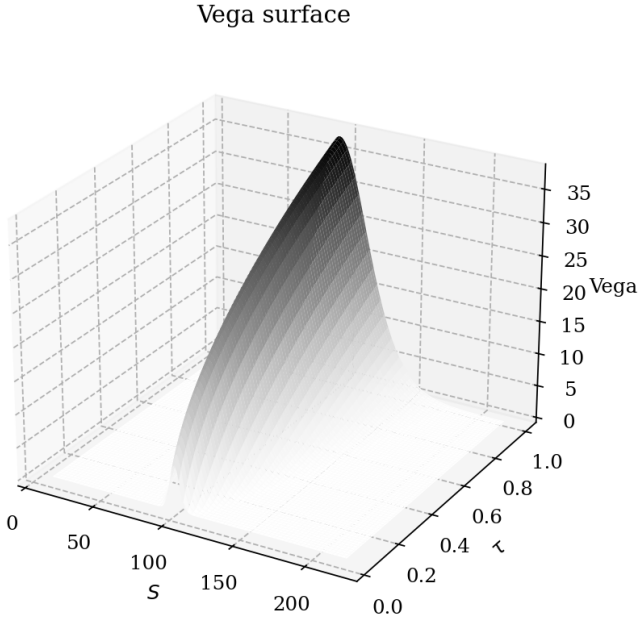


Figure 9: The option Vega, $\partial C/\partial\sigma = S\phi(d_1)\sqrt{\tau}$, as a surface in (S, τ). Vega is maximal near at-the-money strikes and vanishes both deep in- and out-of-the-money, and grows with $\sqrt{\tau}$ for fixed moneyness.

9 Extensions

9.1 Continuous dividends

If the underlying pays a continuous dividend yield q, the risk-neutral drift of S_t becomes $r - q$ rather than r, and (11) acquires an extra term $-qS\,\partial_S V$. The same three-step reduction of Section 5 applies verbatim with r replaced by $r - q$ in the drift terms (Step 1 and the definition of α) while r is retained in the discounting (Step 3, via β). The resulting formula is the Merton (1973) dividend-adjusted Black–Scholes formula, $C = Se^{-q\tau}N(d_1) - Ke^{-r\tau}N(d_2)$, with $d_{1,2}$ using $r - q$ in place of r.

9.2 Time-dependent (deterministic) volatility

If $\sigma = \sigma(t)$ is deterministic, Step 1 of Section 5 still linearizes the spatial operator, but the coefficients of (16) now depend on τ. The substitution (17) generalizes by replacing $\beta\tau$ with $\int_0^\tau \beta(s)\,\mathrm{d}s$, and the resulting heat equation has a time-dependent diffusivity $\frac{1}{2}\sigma(\tau)^2$; the kernel (3) generalizes by replacing $2\alpha t$ with $\int_0^t \sigma(s)^2\,\mathrm{d}s$, i.e. realized variance replaces $\sigma^2\tau$ everywhere in (30).

9.3 Other underlyings

Options on currencies (Garman–Kohlhagen) and on futures (Black's model) fit the same template with the drift term of Step 1 modified ($r - r_f$ for FX, 0 for futures under the appropriate forward measure), leaving the diffusion term, and hence the reduction to the heat equation, unchanged.

9.4 Beyond constant volatility

Once σ is allowed to depend on S (local volatility) or to be itself stochastic, the second-order coefficient of (1) is no longer of Cauchy–Euler form after the log-substitution, and the reduction

to the *constant-coefficient* heat equation of Section 5 generally fails. One still obtains a linear parabolic PDE with a Feynman–Kac representation, but its fundamental solution is no longer the elementary Gaussian kernel (3), and closed-form solutions are the exception rather than the rule.

10 Discussion

10.1 Limits of the physical analogy

The analogy is exact at the level of the PDE, but the boundary and initial conditions have opposite meaning: in the heat equation, $u(x, 0)$ is a physical initial temperature profile evolving forward in time; in Black–Scholes, $\Phi(S)$ is a terminal payoff, and the equation must be read backward in t (equivalently, forward in $\tau = T - t$, as done throughout Section 5). The analogy also does not extend to American options, whose early-exercise feature turns (1) into a free-boundary (obstacle) problem with no elementary heat-kernel solution.

10.2 Risk-neutral measure versus physical diffusion

In physics, Γ literally describes the diffusion of heat, a physically observable process. In finance, the analogous diffusion (of $\ln S_t$ under $\mathbb{Q}$) is a mathematical device: $\mathbb{Q}$ is not the measure governing the observed evolution of S_t (which is $\mathbb{P}$, with drift $\mu \neq r$ in general). The heat equation reduction should therefore be understood as a statement about the *pricing operator*, not about the physical dynamics of the market.

10.3 Numerical consequences

The reduction of Section 5 is also the theoretical justification for the Crank–Nicolson finite-difference scheme commonly used to price options numerically: discretizing the constant-coefficient heat equation (21) on a uniform grid in (x, τ) is markedly more stable and accurate than discretizing (11) directly in (S, t), precisely because the transformed equation has removed the S- and S^2-dependent coefficients that would otherwise degrade the scheme's accuracy near $S \to 0$.

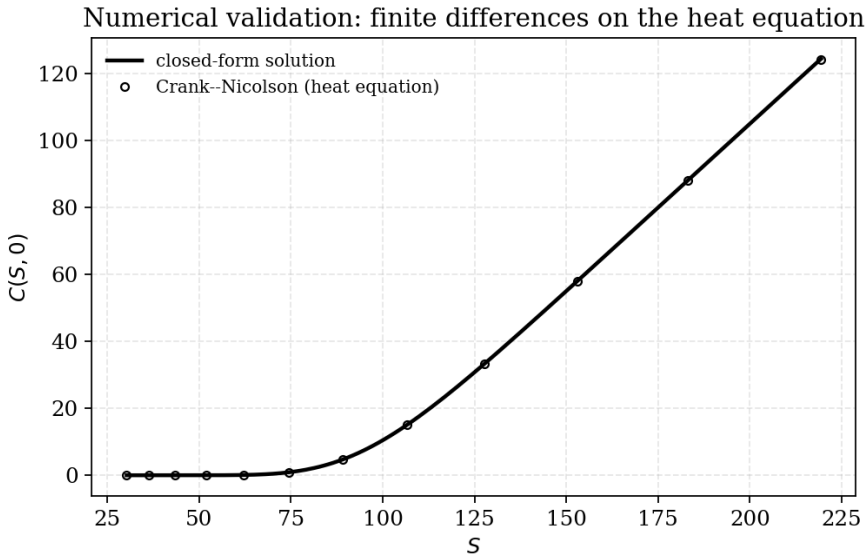


Figure 10: Numerical validation of the transformation: the closed-form Black–Scholes price (solid line) against a Crank–Nicolson finite-difference solution of the heat equation (21), mapped back through the change of variables of Section 5. The two are indistinguishable to plotting accuracy, confirming both the analytic derivation and the practical viability of pricing on the log-price grid.

11 Conclusions

We have shown, through an explicit and fully worked-out sequence of changes of variable — logarithmic in space, reversal in time, exponential tilt — that the Black–Scholes PDE is the heat equation in disguise. Solving the heat equation via its fundamental Gaussian solution and undoing the substitutions reproduces the closed-form Black–Scholes formula without recourse to martingale arguments, Girsanov's theorem, or an explicit choice of the risk-neutral measure — all of that content is, in this derivation, encoded implicitly in the algebra of Section 5. The exercise clarifies why the Black–Scholes formula involves the normal CDF (it is a tail probability of the heat kernel), why Gamma and Theta trade off against each other (they are u_{xx} and u_τ), and why finite-difference pricers are built on log-price grids (that is where the PDE becomes the heat equation).

A The heat kernel: full derivation

We derive (3) by similarity reduction. Seek $u(x,t) = t^{-1/2}F(\xi)$, $\xi = x/\sqrt{t}$ (taking $x_0 = 0$ without loss of generality). Then

$$u_t = -\tfrac{1}{2}t^{-3/2}F(\xi) - \tfrac{1}{2}t^{-3/2}\xi F'(\xi), \qquad u_{xx} = t^{-3/2}F''(\xi).$$

Substituting into $u_t = \alpha u_{xx}$ and multiplying by $t^{3/2}$:

$$-\tfrac{1}{2}F - \tfrac{1}{2}\xi F' = \alpha F'' \quad \Longleftrightarrow \quad \alpha F'' + \tfrac{1}{2}\xi F' + \tfrac{1}{2}F = 0.$$

Noting that $\frac{1}{2}\xi F' + \frac{1}{2}F = \frac{1}{2}(\xi F)'$, integrate once: $\alpha F' + \frac{1}{2}\xi F = c_1$. Requiring decay at infinity forces $c_1 = 0$, so $F'/F = -\xi/(2\alpha)$, whence $F(\xi) = c_2 \exp(-\xi^2/4\alpha)$. Fixing c_2 by $\int_{\mathbb{R}} u(x,t)\,\mathrm{d}x = \int_{\mathbb{R}} F(\xi)\,\mathrm{d}\xi = 1$ (using $\int_{\mathbb{R}} e^{-\xi^2/4\alpha}\mathrm{d}\xi = \sqrt{4\pi\alpha}$) gives $c_2 = 1/\sqrt{4\pi\alpha}$, reproducing (3).

B The convolution integral: intermediate steps

We fill in the completion of the square used in Section 6. For $I(\lambda) = \int_a^\infty (2\pi\sigma^2\tau)^{-1/2} e^{\lambda y - (x-y)^2/2\sigma^2\tau} \mathrm{d}y$, expand

$$\lambda y - \frac{(x-y)^2}{2\sigma^2\tau} = \lambda y - \frac{x^2 - 2xy + y^2}{2\sigma^2\tau} = -\frac{1}{2\sigma^2\tau}\Big(y^2 - 2y(x + \lambda\sigma^2\tau)\Big) - \frac{x^2}{2\sigma^2\tau}.$$

Completing the square in y on the first term, $y^2 - 2ym = (y-m)^2 - m^2$ with $m = x + \lambda\sigma^2\tau$, gives

$$= -\frac{(y-m)^2}{2\sigma^2\tau} + \frac{m^2 - x^2}{2\sigma^2\tau} = -\frac{(y-m)^2}{2\sigma^2\tau} + \lambda x + \frac{\lambda^2\sigma^2\tau}{2},$$

using $m^2 - x^2 = (m-x)(m+x) = \lambda\sigma^2\tau(2x + \lambda\sigma^2\tau)$, divided by $2\sigma^2\tau$, which simplifies to $\lambda x + \lambda^2\sigma^2\tau/2$. This reproduces the exponent used in (25), and the remaining Gaussian integral over (a, ∞) is, by definition of the standard normal CDF and a shift $z = (y-m)/\sqrt{\sigma^2\tau}$, equal to $N\big((m-a)/\sigma\sqrt{\tau}\big)$.

C The Feynman–Kac theorem

Teorema 1 (Feynman–Kac)**.** *Let X_t solve $\mathrm{d}X_t = \mu(X_t, t)\,\mathrm{d}t + \sigma(X_t, t)\,\mathrm{d}W_t$ and let $V(x,t) := \mathbb{E}\big[e^{-\int_t^T r(X_s,s)\mathrm{d}s} f(X_T) \mid X_t = x\big]$. Then, under regularity conditions on μ, σ, r, f, V solves*

$$\frac{\partial V}{\partial t} + \mu(x,t)\frac{\partial V}{\partial x} + \frac{1}{2}\sigma(x,t)^2\frac{\partial^2 V}{\partial x^2} - r(x,t)V = 0, \qquad V(x,T) = f(x).$$

Applying this with $X_t = S_t$ under $\mathbb{Q}$ ($\mu = rS$, $\sigma = \sigma S$, $r(x,t) \equiv r$) reproduces (11), confirming the equivalence, noted in Section 7, between the PDE route of this article and the probabilistic route based on risk-neutral expectations.

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